Balanced forces can go with steady motion

GUIDED NOTES / use with the projected diagram Name _____

A SAY THE WORD

Inertia / ES: inercia / PT: inércia

More mass resists velocity change more.

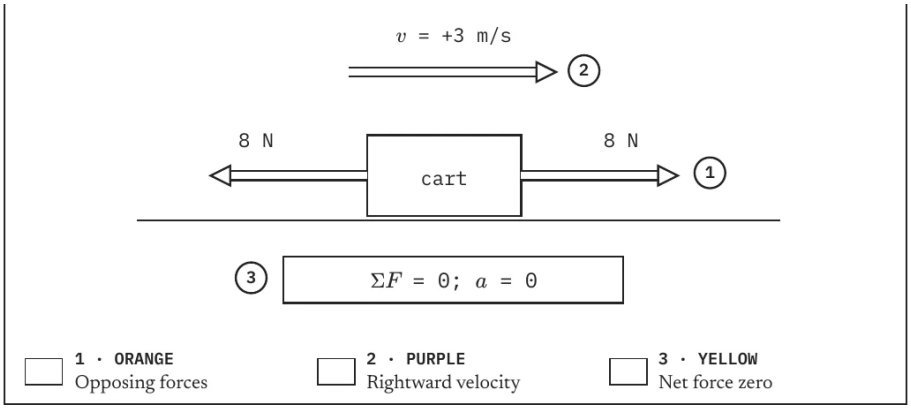
Also on your note page: Inertial frame, Net force, Equilibrium.

B READ THE RELATIONSHIP

Balanced forces → zero acceleration → constant velocity

As you read, find: balanced · steady · changes.

C TRACE THE ORIGINAL FIGURE



Trace equal forces first. Then trace velocity. In pencil, label the arrows F or v.

D FOLLOW THE WORKED EXAMPLE

A cart is at −2 m , moving at +3 m/s . Its external forces balance for the next 4 s . Find its final position.

Given x 0 = −2 m v i = +3 m/s Σ F = 0; t = 4 s

Rule First law: constant v . x f = x 0 + vt .

Work a = 0 m/s² x f = −2 + 3(4)

Answer +10 m Net force zero; motion continues.

E SAVE FOR THE FINAL MEETING

XJ is moving. A forward push is not needed to keep it moving. Gravity can still change its velocity.

Read the physics: Inertia

SUPPORTED READING / read one part, then point to the evidence Name _____

01 PICTURE IT

Push a cart across a rough floor, then let go. It slows because friction still acts on it. The floor and the cart interact. Inertia is a property of the cart; it does not push.

02 HOW IT WORKS

A force is a push or pull. Net force means all the forces on one object added with their directions. When the forces balance, acceleration is zero. A moving cart keeps its speed and direction. A resting cart stays at rest.

03 WHAT TO CHECK

Use a frame that is not speeding up or turning. A purple velocity arrow shows motion. Orange arrows show forces. Never add the purple arrow to the orange arrows.

04 TRY THE EXAMPLE FROM YOUR NOTES

A cart is at −2 m , moving at +3 m/s . Its external forces balance for the next 4 s . Find its final position. +10 m Net force zero; motion continues.

Model check: Balanced forces → zero acceleration → constant velocity.

05 CONNECT TO XJ

XJ needs no engine to keep moving. Over a short interval, use steady velocity as a model. A gravity pull can change the speed or direction. A new observation can test the model.

SAY IT BEFORE YOU WRITE

Is the cart moving, or is its velocity changing? How can you tell? The cart moves ___. Its velocity ___ because the forces ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Inertial frame

ES: sistema de referencia inercial / PT: referencial inercial

Meaning in my words

How would you explain inertial frame to a friend?

In my words, this means ___.

How can you tell?

A puck glides straight at steady speed. Would a camera moving straight at constant speed also see steady motion?

I can tell because ___.

Example or sketch

Which bus ride could serve as an inertial frame for watching a puck: cruising straight or turning?

My example shows ___.

Non-example and why

A bus brakes and a loose puck seems to surge forward inside it. Does that prove a forward force acted on the puck?

I would change ___ because ___.

Net force

ES: fuerza neta / PT: força resultante

Meaning in my words

How would you explain net force to a friend?

In my words, this means ___.

How can you tell?

Two people push one cart in opposite directions. Why do the arrow directions matter?

I can tell because ___.

Example or sketch

One person pushes 6 N right and another 4 N left. What single arrow replaces both?

My example shows ___.

Non-example and why

A student adds those pushes to get 10 N right. Which push did they count the wrong way?

I would change ___ because ___.

Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

Inertia

ES: inercia / PT: inércia

Meaning in my words	How can you tell?
What does inertia mean? Say it the way you'd explain it to a friend. In my own words, inertia means …	An empty cart and a loaded cart both start at rest. We push each with the same net force for one second. Which gains less speed? Why? The ___ cart gains less speed because …
Example or sketch	**Non-example and why**
A cart is already rolling right. The forces on it balance. Where would you draw it one second later? And one second after that? Label your three carts 0 s, 1 s and 2 s.	Someone points to the hand pushing the cart and says, "That push is inertia." What would you tell them? The hand supplies ___. The cart has ___.

Equilibrium

ES: equilibrio / PT: equilíbrio

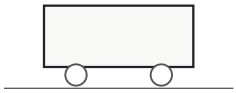
Meaning in my words	How can you tell?
How would you explain equilibrium to a friend? In my words, this means ___.	A cart rolls straight at steady speed. What must the horizontal forces add up to? I can tell because ___.
Example or sketch	**Non-example and why**
Sketch the horizontal forces on a cart with a 3 N pull right and 3 N friction left. My example shows ___.	A tossed ball is briefly still at the top. Is it in equilibrium there? I would change ___ because ___.

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw a moving cart with balanced force arrows in orange and its separate velocity arrow in purple.

Labels you can use: balanced · steady · changes.



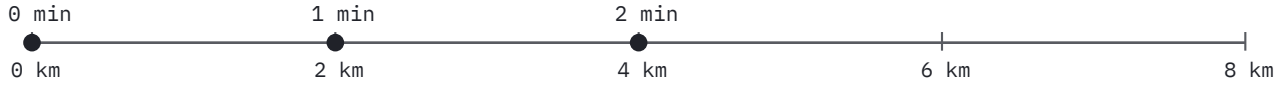
Tell a partner

The cart moves ___. Its velocity ___ because the forces ___.

Where should we point the telescope?

SECTION E / FINAL MEETING Name _____

The telescope spots XJ once a minute. At minute 3, the camera can take one picture. Your team gets to choose where it points.



Camera windows A +4 km B +6 km C +8 km

Choose a job together. One of you marks the prediction. The other makes the call.

☐ **Tracking engineer**

Circle the window you'd aim for. Tell the camera operator where to point and which two sightings helped you decide.

☐ **Live science broadcaster**

Tell viewers where to look. Someone asks, "If XJ has no engine, why does it keep moving?" Use your cart drawing to answer.

Try saying "Aim at ___ because ___." Or start your broadcast with "Watch window ___."

Mark your prediction here.



WHEN THE NEXT SIGHTING COMES IN

Mark the new sighting. How far did XJ move in the last minute? Switch speakers and tell your crew or your viewers what changed.

"We predicted ___. The telescope saw ___. In the last minute, XJ moved ___."

Why could a straight-line forecast work for a minute even while gravity pulls on XJ?